

MINGYUAN JIA

✉ jmy23@mails.tsinghua.edu.cn · ☎ (+86)137-1822-0153 · [Google Scholar](#)

EDUCATION

Tsinghua University, *Department of Automation*

Sep. 2023 - *Present*

- **Talent Program for Intelligent Robotics**

- **Core Courses:** Major GPA: 3.7, Overall: 3.88 (Top 20%); Achieved A-/A in core courses including Fundamentals of Analog Electronics, Signals and Systems, Modern Electronic System Design, Pattern Recognition and Machine Learning, Operations Research, and University Physics.

SKILLS & INTERESTS

- **Research Interests:** World Model in Embodied System, Robotics, Representation Learning.
- **Technical Skills:** Proficient in Python, C/C++, ROS, and PyTorch deep learning framework. Experienced in model training and inference based on Transformers and Diffusion architectures. Skilled in Linux, Shell, Git, and modern development workflows.
- **Languages:** Proficient in English academic reading and writing. Capable of fluent daily and technical communication.
- **Competencies:** Highly self-driven with rapid learning capabilities; excels in team collaboration and communication.

PUBLICATIONS

* denotes co-first author; † denotes project leader.

AirScape: An Aerial Generative World Model with Motion Controllability

2025

Baining Zhao*, Rongze Tang*, **Mingyuan Jia***, Ziyu Wang, Fanghang Man, Xin Zhang, Yu Shang, Weichen Zhang, Wei Wu, Chen Gao, Xinlei Chen, Yong Li.

- **ACMMM 2025 BNI Oral.** Co-first author of AirScape –a motion-controllable generative world model for urban UAVs. Designed data curation and self-play training framework, built a high-quality aerial dataset, contributed to experiments, paper revision, and open-source maintenance.

Efficient Real-Time Interactive World Modeling with Versatile Control

2026

Yu Meng*, Chen Gao†, Baining Zhao*, **Mingyuan Jia***, Rongze Tang, Ziyu Wang, Jianjie Fang, Wan Qin, Yingshan Lei, Jiaao Tang, Jiaao Li, Xin Zhang, Yu Shang, Wei Wu, Xinlei Chen, Xiao-Ping Zhang, Yong Li.

- **Submitted to ICML 2026.** Co-first author of WorldScape –an autoregressive video diffusion model with real-time streaming generation (16 FPS on H100) via asymmetric distillation. Designed **3D Spatial Consistency Training** and **Memory-Oriented KV Cache Optimization** to ensure physical consistency and spatial coherence in long-horizon generation.

Worldscape-MoE: A Unified Mixture-of-Experts World Model for Scalable Heterogeneous Action Control

2026

Jianjie Fang†*, Yongyan Xu*, Ziyu Wang*, Yuchao Huang, Zhaolu Wang, Rongze Tang, **Mingyuan Jia**, Baining Zhao, Weichen Zhang, Xin Zhang, Haisheng Su, Yu Shang, Chen Gao, Wei Wu, Xinlei Chen, Yong Li

- **Submitted to NeurIPS 2026.** Proposed a framework unifying heterogeneous control signals via MoE-enhanced DiT. Using shared expert for cross-control world knowledge and modality-specific experts for specialization, with a progressive tuning strategy for continual expansion.

WorldREPA: Extending Representation Alignment for Multi-source World Knowledge Learning

2026

Jiaao Tang*, **Mingyuan Jia**†*, Zhaolu Wang*, Xinhao Jin, Xin Zhang, Jianjie Fang, Baining Zhao, Yu Shang, Chen Gao, Wei Wu, Yong Li.

- **Submitted to NeurIPS 2026.** Project leader of WorldREPA, a framework for multi-source alignment of DiT. Proposed lightweight injection of geometry & semantics, and designed representation analysis with core modules for consistency and fidelity.

ReplicateAnyScene: Zero-Shot Video-to-3D Composition via Textual-Visual-Spatial Alignment 2026

Mingyu Dong^{}, Chong Xia^{*}, **Mingyuan Jia^{*}**, Weichen Lyu, Long Xu, Zheng Zhu, Yueqi Duan.*

- **Submitted to 3DV 2027.** Co-first author of ReplicateAnyScene – a zero-shot framework for compositional 3D scene reconstruction from videos without manual prompts or auxiliary inputs. Built the C3DR benchmark with unified baseline replication and multi-dimensional metrics for systematic evaluation.

INDUSTRY EXPERIENCE

Beijing Manifold AI Technology Co., Ltd.

Nov. 2025 - Apr. 2026

R&D Intern Mentors: Chen Gao, Wei Wu

- Key contributor to WorldScape model. Secured **1st place** on WorldScore Leaderboard.

Shanghai Coco Matrix Intelligent Technology Co., Ltd.

Jul. 2026 - Present

R&D Intern Mentors: Yuxiang Gao

- One of the core members of the robotic model pre-training team.

ACADEMIC EXPERIENCE

FIBLAB,THU

May. 2025 - Apr. 2026

Advised by:Prof. Yong Li

Tsinghua Spatial Intelligence & Vision Group,THU

Feb. 2026 - Aug. 2026

Advised by:Prof. Yueqi Duan

HONORS & ACTIVITIES

- **Grand Prize**, 43rd Tsinghua University "Challenge Cup" AI TA Challenge: Responsible for Dify workflow construction, RAG knowledge base development, and end-to-end API integration and optimization, supporting stable system deployment and high-quality live demonstrations.
- **Second Prize**, Robot Dog Development Competition (Two editions): Developed vision capture and motion control modules based on ROS and RGB-D cameras. Implemented dynamic pose adjustment through host communication, significantly improving movement stability during ball-tracking tasks.
- **Second Prize**, Embodied AI Mini-Car Challenge: Led dataset construction and model training for the vision system. Developed the target recognition pipeline and coordinated with a robotic arm for autonomous grasping, improving task success rates and real-time performance.
- **164 Hours of Volunteer Service**: Actively participated in class and university-organized volunteering activities, including university sports meets and faculty events.